Poisson's and Laplace's equations

- We have learned $\nabla \cdot \mathbf{D} = \rho$
  - In a linear and isotropic medium $\mathbf{D} = \varepsilon \mathbf{E}$
  - Moreover, $\mathbf{E} = -\nabla V$
  - Therefore

$$\nabla \cdot (\nabla V) = \nabla^2 V = -\frac{\rho}{\varepsilon}$$

- Poisson's equation

$$\nabla^2 V = -\frac{\rho}{\varepsilon}$$

- At points in a simple medium where there is no free charge

$$\nabla^2 V = 0 \text{ ----- Laplace's equation}$$

Example

The two plates of a parallel-plate capacitor are separated by a distance d and maintained at potentials 0 and V_0. Assuming negligible fringing effect at the edges, determine (a) the potential at any point between the plates, and (b) the surface charge densities on the plates

Solution:

– There is no free charges inside the medium between plates

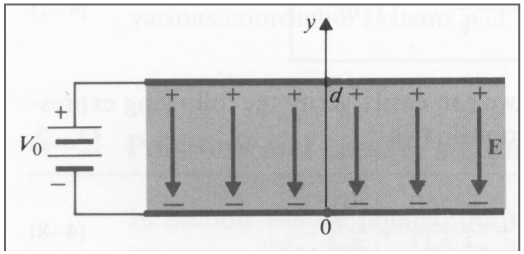
$$\nabla^2 V = 0$$

– There is no variation of V in x directions, V is function of y only

$$\nabla^2 V = \frac{dV^2}{dy^2} = 0 \Rightarrow V(y) = C_1 y + C_2$$

– Applying boundary conditions: V=0 at y=0; V=V_0 at y=d

$$V(y) = \frac{V_0}{d} y$$



Example (cont'd.)

$$\mathbf{E} = -\nabla V = -\frac{\partial V}{\partial x}\mathbf{a}_x - \frac{\partial V}{\partial y}\mathbf{a}_y - \frac{\partial V}{\partial z}\mathbf{a}_z = -\mathbf{a}_y \frac{V_0}{d}$$

– From lecture 6

$$\left.\begin{aligned} \mathbf{E} &= -\mathbf{a}_y \frac{\rho_s}{\varepsilon} \\ \mathbf{E} &= -\mathbf{a}_y \frac{V_0}{d} \end{aligned}\right\} \rho_s = \frac{\varepsilon V_0}{d}$$

– Surface charge density is positive at the upper plate, negative at the lower plate

Electrical properties of materials

- Constitutive parameters of a material: ε, μ, σ
- A material is
  - Homogeneous, if the constitutive parameters are constant
  - Isotropic, if the constitutive parameters are independent of direction
- Materials can be classified into three types
  - Conductors (most metals) $\sigma >> 1$
    - Free electrons
  - Insulators (or dielectrics) $\sigma << 1$
    - Electrons are bounded, cannot move freely
  - Semiconductors
    - Fall between conductors and dielectrics
    - Possess a relatively small number of freely movable charges

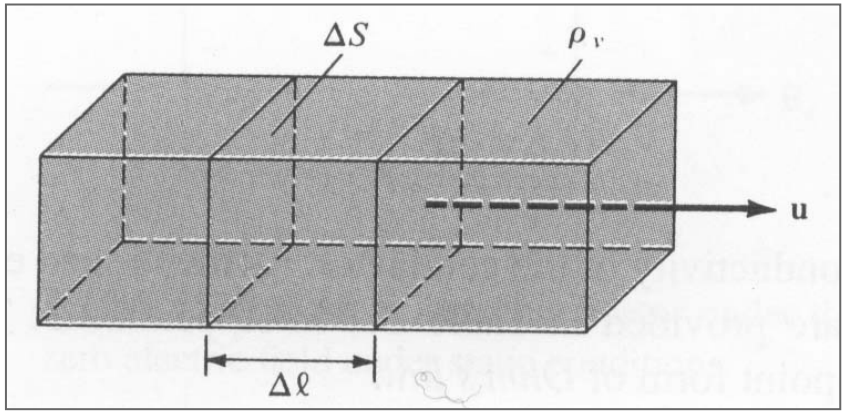
Current density and Ohm's law

- The current through a given area is the electric charge passing through the area per unit time

$$I = \frac{dQ}{dt}$$

- If there is a flow of charge, with density ρ, at velocity **u**, flowing through an area Δs

  Charge passing Δs in time dt



$$dQ = \rho dV = \rho \Delta s \Delta l = \rho \Delta s \frac{\Delta l}{dt} dt = \rho \mathbf{u} \cdot \mathbf{a}_n \Delta s dt$$

Current density

- Current through area Δs

$$\Delta I = \frac{dQ}{dt} = \rho \mathbf{u} \cdot \mathbf{a}_n \Delta s$$

- The **current density** at a given point is the current through a unit normal area at that point

$$\mathbf{J} = \rho \mathbf{u} \ \ (\mathrm{A/m^2})$$

$$\Delta I = \mathbf{J} \cdot \Delta \mathbf{s} \Rightarrow I = \int_S \mathbf{J} \cdot d\mathbf{s}$$

Conductivity

- When an external field **E** is applied
  - Electrons will move with an average drift velocity $\mathbf{u}_e$
    - $\mathbf{u}_e$ is proportional to **E**

$$\mathbf{u}_e = -\mu_e \mathbf{E}$$

    - μ_e is the **electron mobility** measured in ($m^2/V \cdot s$)
    - Negative sign because electron moves against **E**
  - Holes will move with an average drift velocity $\mathbf{u}_h$

$$\mathbf{u}_h = \mu_h \mathbf{E}$$

    - μ_h is the hole mobility
- Therefore

$$\mathbf{J} = \mathbf{J}_e + \mathbf{J}_h = \rho_e \mathbf{u}_e + \rho_h \mathbf{u}_h$$
$$= \left(-\rho_e \mu_e + \rho_h \mu_h\right)\mathbf{E} = \sigma \mathbf{E}$$

$$\sigma = -\rho_e \mu_e + \rho_h \mu_h$$

  - σ is the **conductivity** of the material
  - ρ_e and ρ_h is the volume density of electrons and holes

Ohm's law

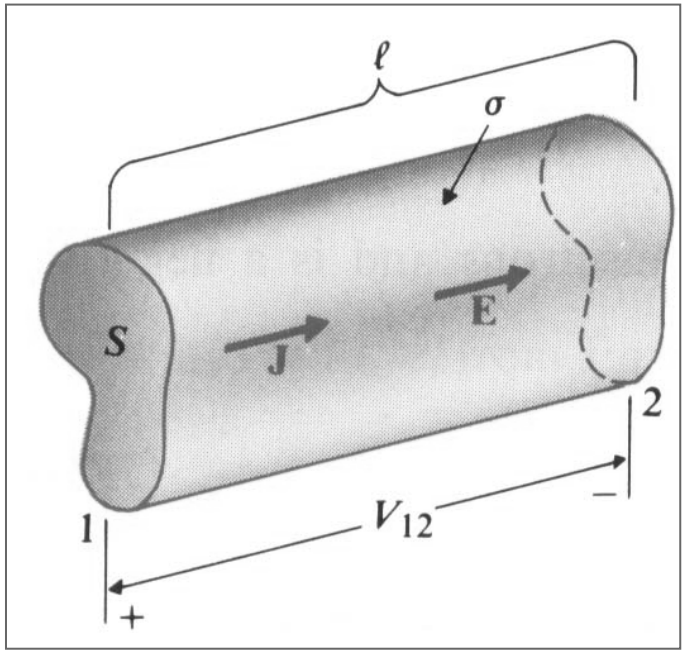
- $\mathbf{J} = \sigma \mathbf{E}$ is the **point form of Ohm's law** .
  - It holds at all points in space
- Within a conductor both **J** and **E** are in the direction of current flow
  - The potential difference $V_{12} = El$
  - Total current

$$I = \int_S \mathbf{J} \cdot d\mathbf{s} = JS$$

Thus

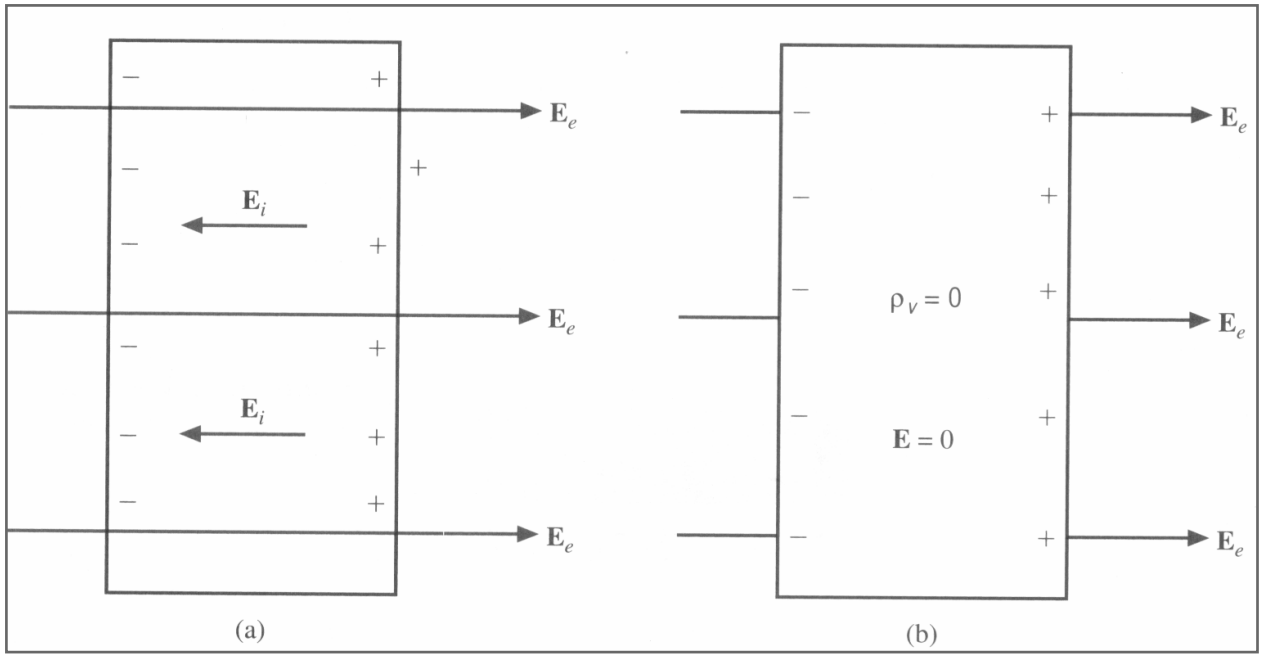
$$J = \frac{I}{S} = \sigma E = \sigma \frac{V_{12}}{l}$$

$$R = \frac{V_{12}}{I} = \frac{l}{\sigma S} \quad (\Omega)$$



Electric field inside a conductor

- In a perfect dielectric, $\sigma = 0$, $\mathbf{J} = 0$ regardless of $\mathbf{E}$
- In a perfect conductor, $\sigma = \infty$, $\mathbf{E} = 0$ regardless of $\mathbf{J}$
- In steady state, the net electric field inside a conductor is 0



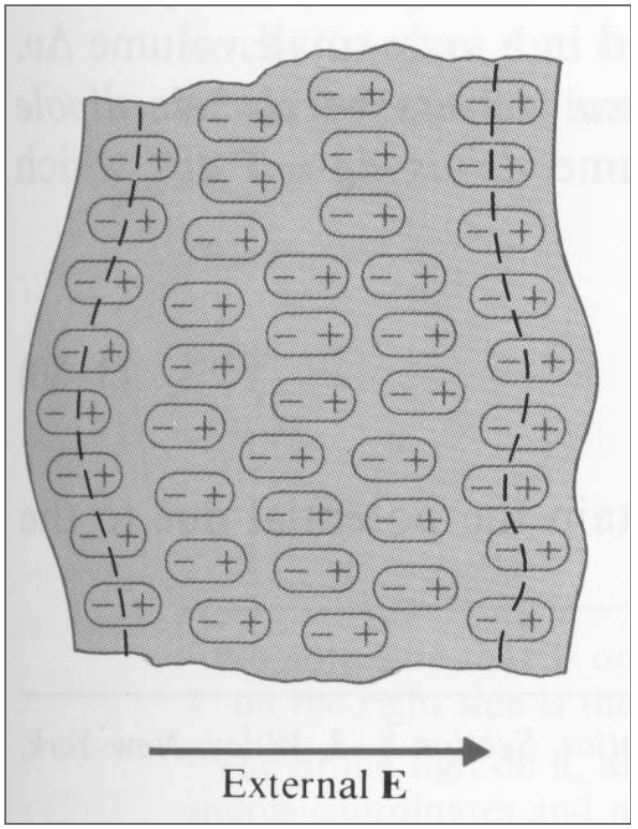
- The electric charge in a conductor can only exist on the surface

$$\mathbf{E} = 0, \qquad \rho_v = 0, \qquad V_{ab} = 0 \qquad \text{inside a conductor}$$

Dielectrics in static electric field

- Ideal dielectric do not contain free charges
- The external electric field causes small displacements of positive and negative charges in opposite directions
  - Electric dipoles are formed
  - Electric dipoles induce non-zero electric field
- A polarization vector is defined to characterize their macroscopic effect

$$\mathbf{P} = \lim_{\Delta v \to 0} \frac{\sum_{k=1}^{n\Delta v} \mathbf{P}_k}{\Delta v} \qquad (\mathrm{C/m^2})$$



Permittivity

- The polarization is directly proportional to **E**

$$\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}$$

where χ_e is a dimensionless quantity called *electric susceptibility*

- The presence of microscopic dipoles alters the relationship between **E** and **D**

$$\begin{aligned}\mathbf{D} &= \varepsilon_0 \mathbf{E} + \mathbf{P} = \varepsilon_0 (1 + \chi_e)\mathbf{E} \\ &= \varepsilon_r \varepsilon_0 \mathbf{E} = \varepsilon \mathbf{E}\end{aligned}$$

where

$$\varepsilon_r = 1 + \chi_e = \frac{\varepsilon}{\varepsilon_0}$$

is a dimensionless quantity known as the relative permittivity (dielectric constant) of the medium. ε is the absolute permittivity

- Thus **D** is also know historically as electric displacement